

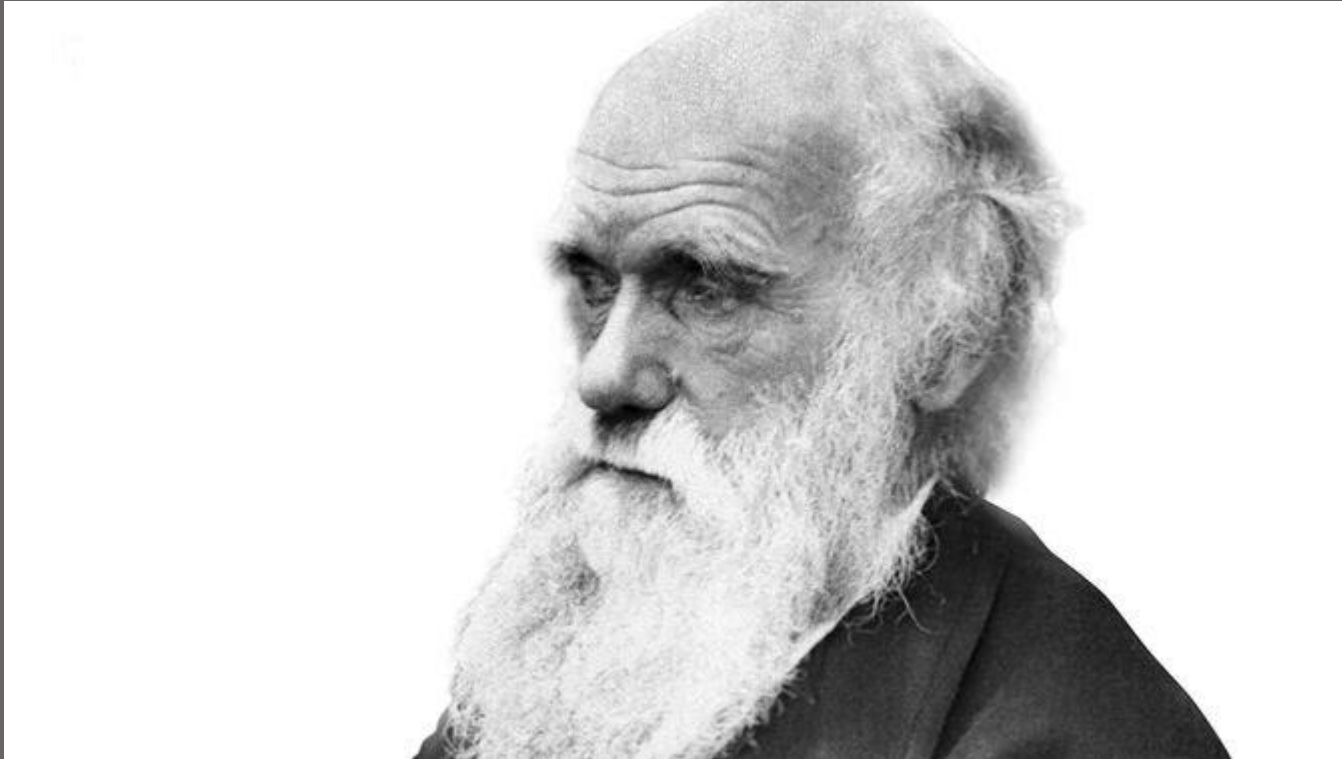


CYCLE- DELIC

Justin Pfeiffer

The Problem

The “Survival of the Fittest”



<https://www.zoo-rosrock.de/darwineum/charles-darwin.html>

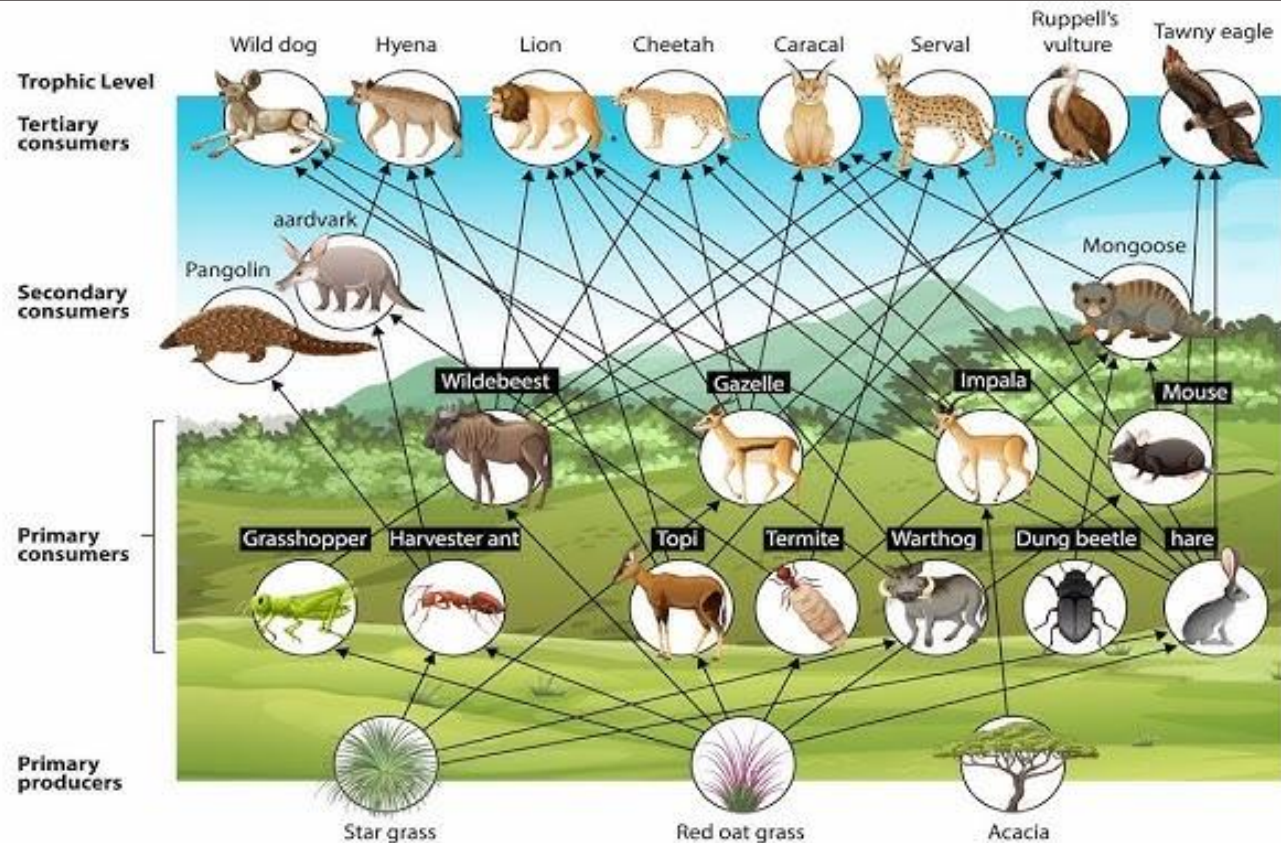
- ◇ If only the fittest survive, why is there so much diversity?
- ◇ There cannot be so many “fittest species”
- ◇ How is biodiversity achieved then?

➡ Via interactions!

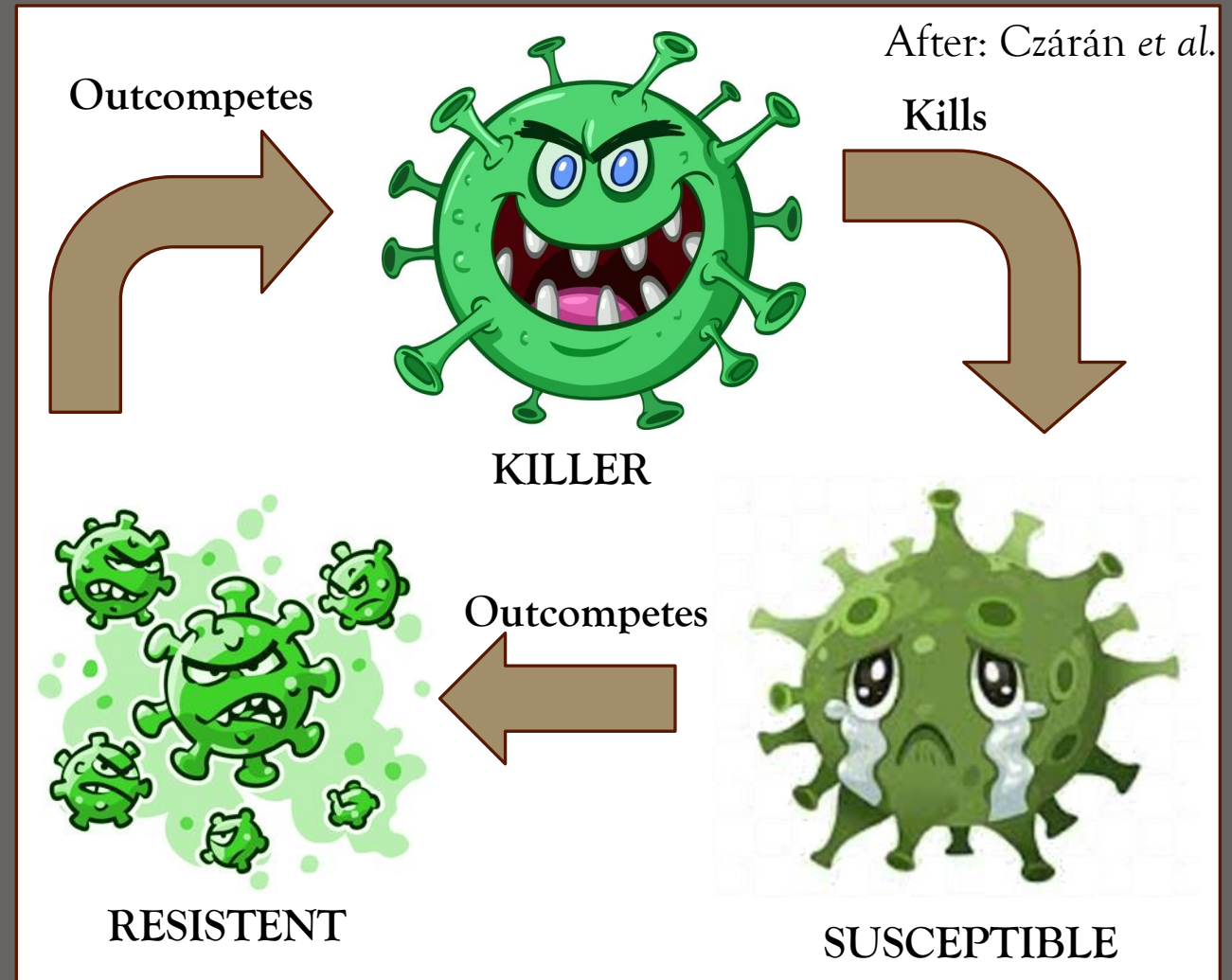
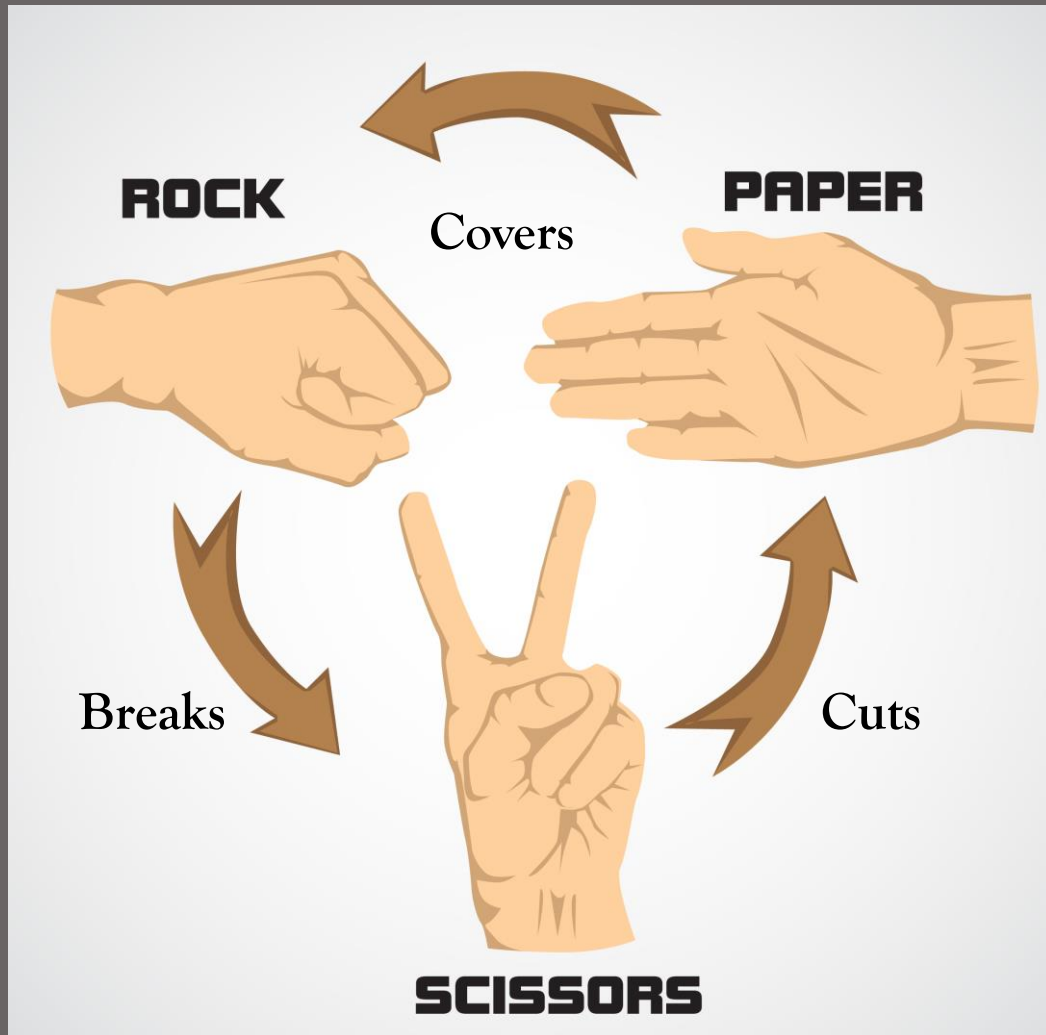
Complicated Systems

Game Theory

- ❖ “Players” are affected by the surrounding players’ actions
- ❖ Here, “players” are populations



Simplified, please



Assumptions

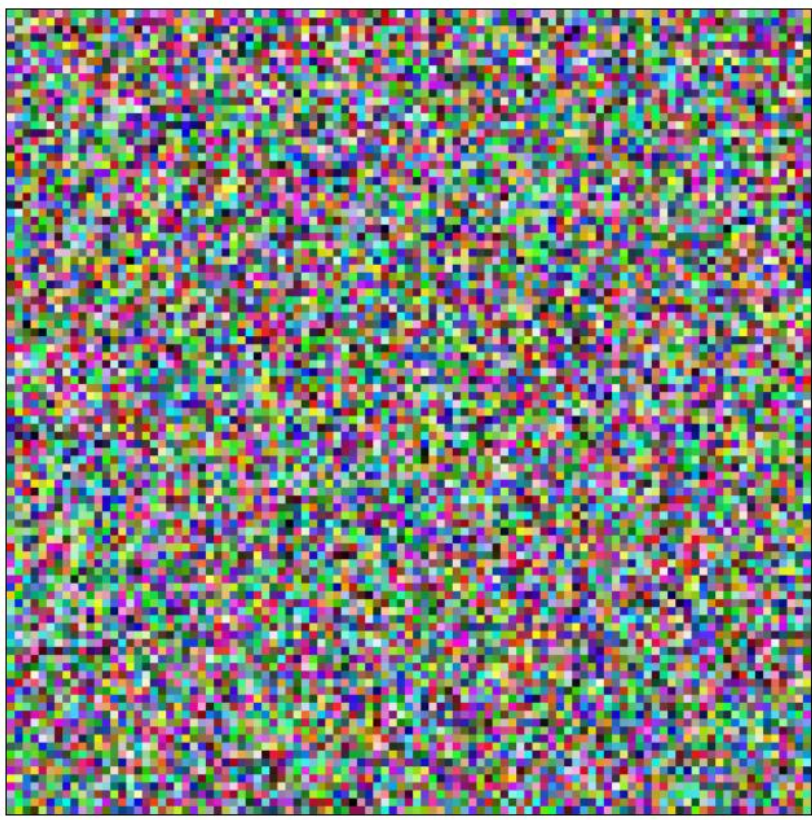


INTERACTION IS COMPETITIVE,
BUT NON-HIERARCHICAL



INTERACTIONS ARE LOCAL

The Model



Individuals on a lattice, randomly distributed.
Red eats Green eats Blue eats Red...

Parameters:

- ◇ Predation: $\longrightarrow$ eating and being eaten
- ◇ Competition: $\longrightarrow$ limited resources for growth
- ◇ Diffusion: $\longrightarrow$ movement through lattice

Interaction is only possible locally

The Patterns

After a short time,
fascinating patterns emerge

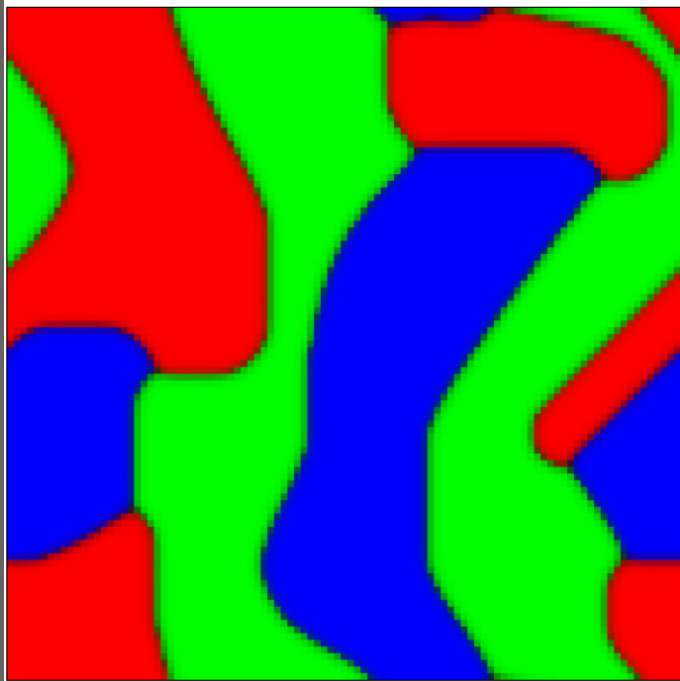
Let's try it!

<https://www.complexity-explorables.org/explorables/cycledelic/>

The Parameters Predation

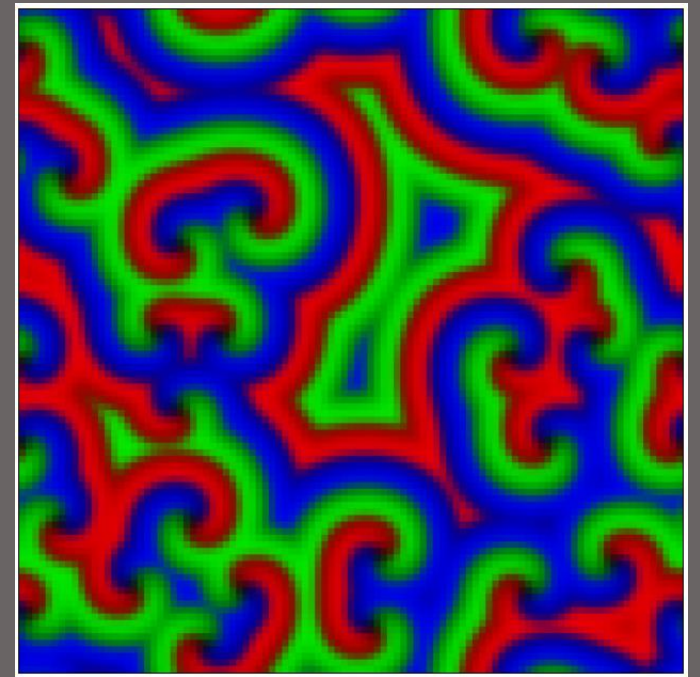
Left: Low
predation

- ◆ Precludes
ordered wave
formation
- ◆ *Very slow*



Right: High
competition

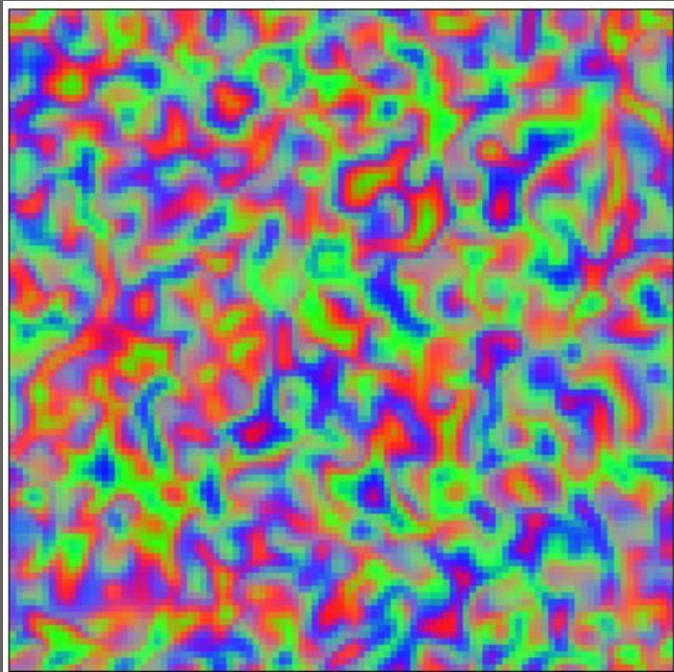
- ◆ Leads to
smaller
waves
- ◆ Movement is
fast



The Parameters: Competition

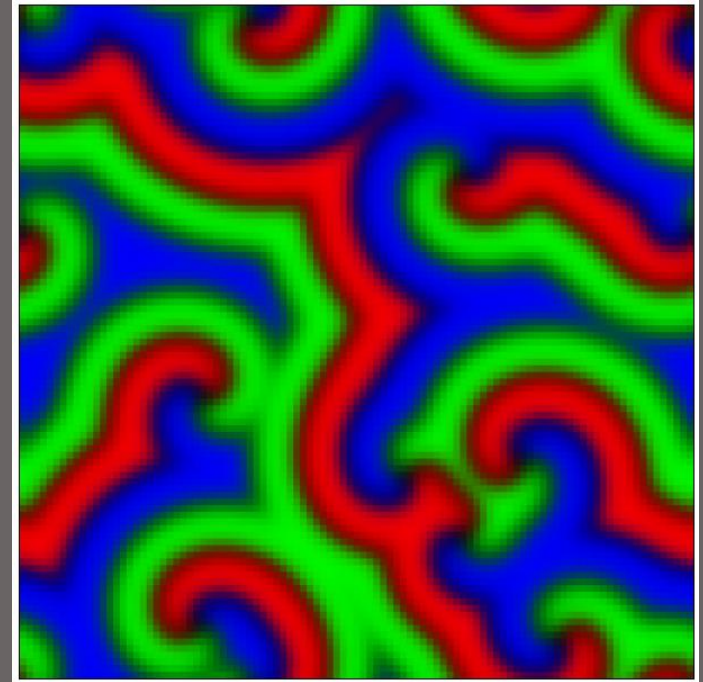
Left: Low
competition

- ◊ Precludes spiral formation
- ◊ Very low competition ends in “whiteout”



Right: High
diffusion

- ◊ Leads to big waves
- ◊ Edges are rougher because of greater speed



The Parameters Diffusion

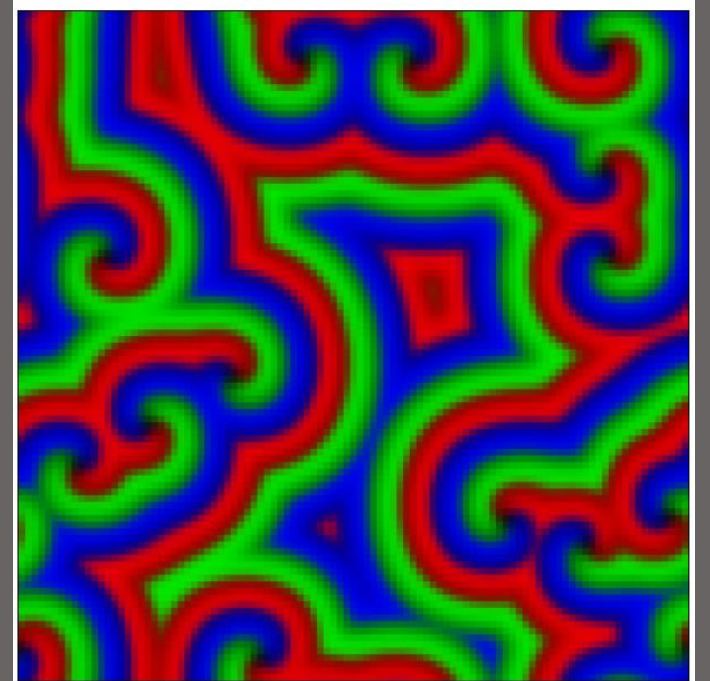
Left: Low
diffusion

- ◇ leads to
small waves



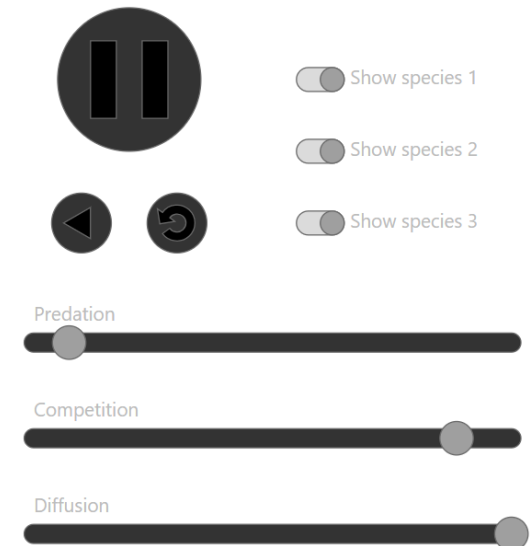
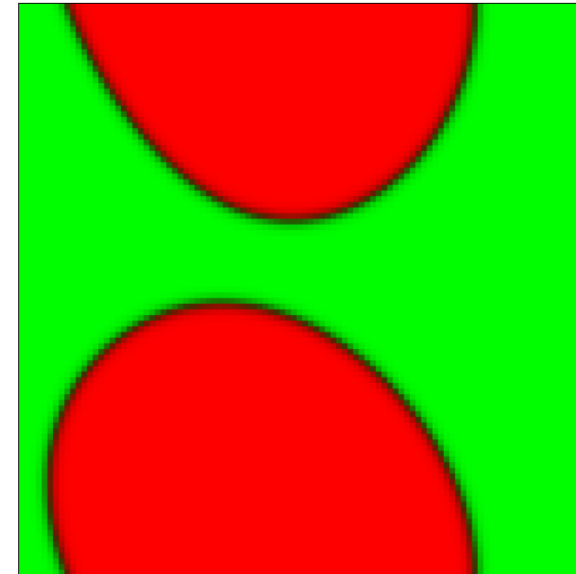
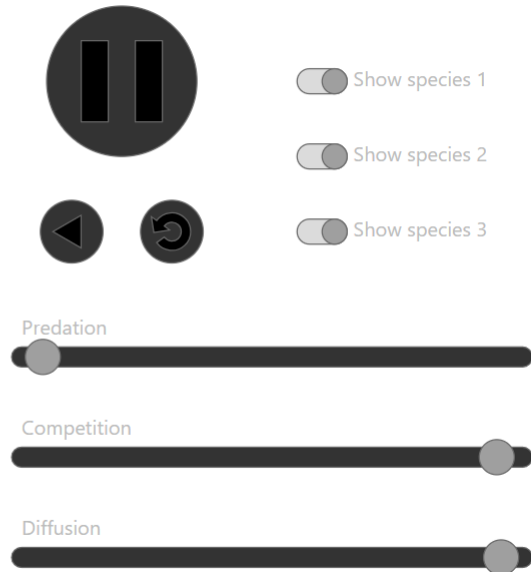
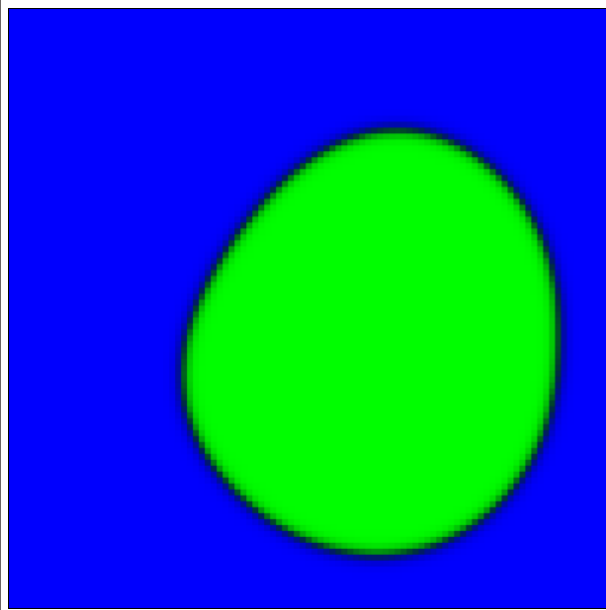
Right: High
predation

- ◇ Leads to
faster and
bigger waves



New Insights

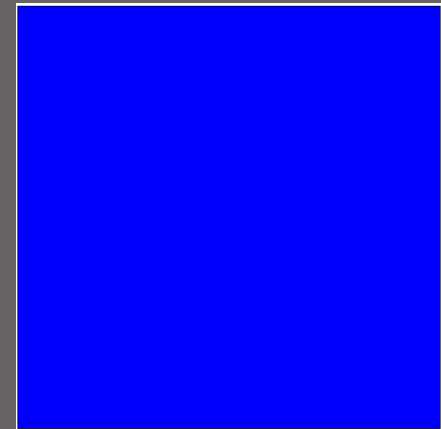
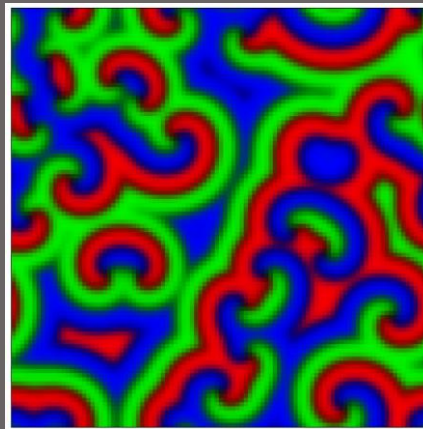
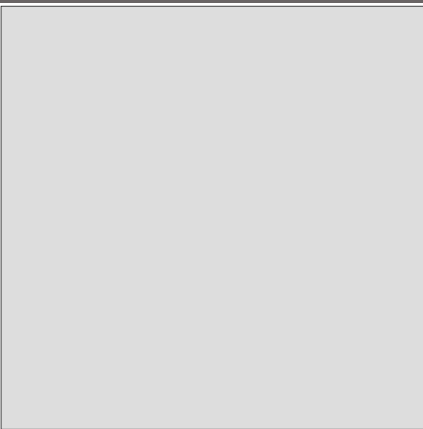
- ◆ The survival of populations is based on the wave patterns
- ◆ If the spirals outgrow the habitat, only one species survives



Results

The model has three main end-states/results:

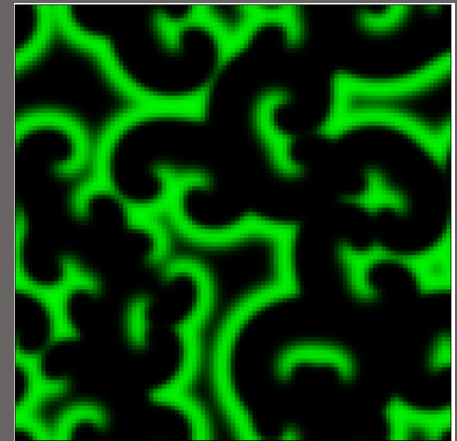
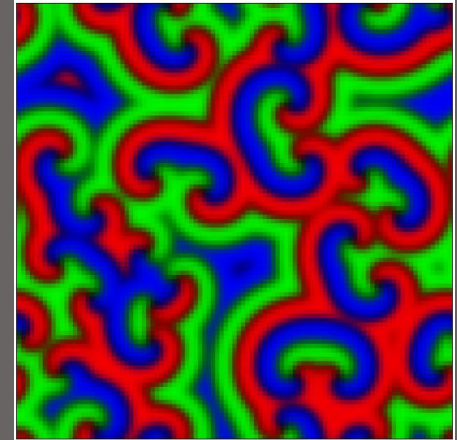
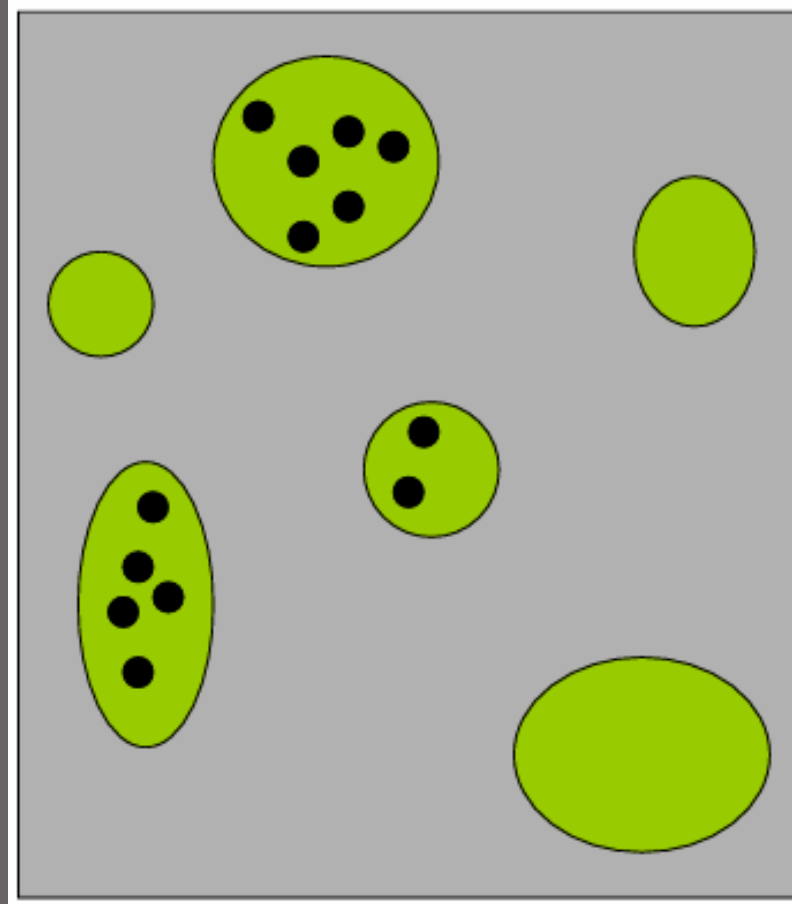
- ◇ “Whiteout” All Species perfectly balanced - only for this model!
- ◇ “Spirals” All Species present, eternally hunting each other
- ◇ “Extinction” Only one species survives, two go extinct



Bigger $\neq$ Better

Metapopulations

- ◇ Single populations are independent
- ◇ One population dying out is tolerable, as the space can be “re-inoculated” by the other populations
- ◇ The system may reach equilibrium again



Sources

- ♦ Christoph Hauert *et al.*, Volunteering as Red Queen Mechanism for Cooperation in Public Goods Games. *Science* **296**, 1129-1132 (2002). DOI: [10.1126/science.1070582](https://doi.org/10.1126/science.1070582)
- ♦ Kerr, B., Riley, M. A., Feldman, M. W., & Bohannan, B. J. M. (2002). Local dispersal promotes biodiversity in a real-life game of rock-paper-scissors. *Nature*, **418**(6894), 171–174. <https://doi.org/10.1038/nature00823>
- ♦ Reichenbach, T., Mobilia, M., & Frey, E. (2007). Mobility promotes and jeopardizes biodiversity in rock-paper-scissors games. *Nature*, **448**(7157), 1046–1049. <https://doi.org/10.1038/nature06095>
- ♦ Siegert, F., & Weijer, C. J. (1995). Spiral and concentric waves organize multicellular *Dictyostelium* mounds. *Current Biology*, **5**(8), 937–943. [https://doi.org/10.1016/S0960-9822\(95\)00184-9](https://doi.org/10.1016/S0960-9822(95)00184-9)
- ♦ T.L. Czárán, R.F. Hoekstra, & L. Pagie, Chemical warfare between microbes promotes biodiversity, *Proc. Natl. Acad. Sci. U.S.A.* **99** (2) 786-790, <https://doi.org/10.1073/pnas.012399899> (2002).